

Filename Mismatch Fix Report

Summary

Root Cause: 30 PNG files in `results/figures/` existed locally but were **not tracked in Git** due to a `.gitignore` rule (`results/figures/*.png`). The original 43 figures had been force-added previously (commit `c9c0f07`), but 30 additional figures generated in later phases were never force-added. This caused **broken image links** when viewing documentation on GitHub.

Fix Applied: Force-added all 30 untracked PNG files to Git tracking (commit `c9406c2`).

Detailed Findings

1. Mismatches Found

No actual filename mismatches exist between documentation references and file names. The earlier concern about `fleet_sensitivity_dual.png` vs `fleet_sensitivity_v2_dual.png` was a false alarm — both files exist as separate figures, and docs correctly reference `fleet_sensitivity_dual.png`.

The real issue was that **30 PNG files were untracked in Git** despite being referenced in documentation.

2. Broken Embedded Images on GitHub (9 files)

These files were referenced via `![alt](/home/ubuntu/ems-optimization/docs/path)` syntax and would render as broken images on GitHub:

File	Referenced In
<code>fleet_sensitivity_dual.png</code>	<code>technical_report.md:472</code> , <code>fleet_sensitivity_dual_investigation.md:17</code>
<code>cbd_scenario_comparison.png</code>	<code>technical_report.md:528</code> , <code>cbd_robustness_analysis.md:81</code>
<code>cbd_response_comparison.png</code>	<code>cbd_robustness_analysis.md:78</code>
<code>cbd_heatmap.png</code>	<code>cbd_robustness_analysis.md:84</code>
<code>queue_comparison_by_policy.png</code>	<code>queue_analysis.md:100</code> , <code>technical_report.md:554</code>
<code>queue_vs_fleet_size.png</code>	<code>queue_analysis.md:103</code>
<code>queue_vs_demand.png</code>	<code>queue_analysis.md:106</code>
<code>queue_heatmap.png</code>	<code>queue_analysis.md:109</code>
<code>seasonal_patterns.png</code>	<code>technical_report.md:575</code>

3. Additional Untracked Files (21 files)

Referenced in `visualization_index.md` , `figure_trace_guide.md` , and other docs via backtick notation:

- `allocation_comparison_K20.png`
- `capacity_sensitivity_heatmap_notebook.png`
- `cbd_robustness.png`
- `demand_sensitivity_curve.png`
- `distance_matrix_coverage.png`
- `fleet_sensitivity_curve.png`
- `p0_spatial_map.png`
- `p0_spatial_metrics.png`
- `p0_spatial_north_south.png`
- `p0_vs_p2_response_time.png`
- `policy_comparison.png`
- `precinct_demand_heatmap.png`
- `precinct_demand_rates.png`
- `precinct_demand_rates_improved.png`
- `production_fleet_sensitivity.png`
- `queue_metrics.png`
- `response_time_coverage_tradeoff.png`
- `seasonal_decomposition.png`
- `seasonal_heatmap.png`
- `statistical_effect_sizes.png`

- `temporal_demand_patterns.png`

4. Action Taken

- **No files renamed** — all filenames already matched their documentation references
- **No documentation updated** — all references were correct
- **30 files force-added to Git** — bypassing `.gitignore` rule, consistent with the original force-add approach in commit `c9c0f07`

5. Post-Fix Verification

- **73 PNG files** now tracked in `results/figures/` (43 original + 30 newly added)
- **All embedded image references** (``) resolve to tracked files
- **All specific backtick references** resolve to tracked files
- **0 broken links** remaining

6. Note on `.gitignore`

The `.gitignore` rule `results/figures/*.png` remains in place. This is intentional — it prevents accidental commits of new generated figures. Critical figures must be explicitly force-added with `git add -f`. This is consistent with the project's approach of only tracking essential deliverables.